In [44]:

```
!pip install pandoc
```

```
Collecting pandoc
  Downloading pandoc-2.4.tar.gz (34 kB)
  Installing build dependencies: started
  Installing build dependencies: finished with status 'done'
  Getting requirements to build wheel: started
  Getting requirements to build wheel: finished with status 'done'
  Preparing metadata (pyproject.toml): started
  Preparing metadata (pyproject.toml): finished with status 'done'
Collecting plumbum (from pandoc)
  Downloading plumbum-2.0.2-py3-none-any.whl.metadata (8.5 kB)
Collecting ply (from pandoc)
  Downloading ply-3.11-py2.py3-none-any.whl.metadata (844 bytes)
Downloading plumbum-2.0.2-py3-none-any.whl (168 kB)
Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
Building wheels for collected packages: pandoc
  Building wheel for pandoc (pyproject.toml): started
  Building wheel for pandoc (pyproject.toml): finished with status 'done'
  Created wheel for pandoc: filename=pandoc-2.4-py3-none-any.whl size=34814 sha256=1d16d
c9a70daa091e38ad70197f9051a54cb91c4a8171dca5f4866792ed3fea0
  Stored in directory: c:\users\dell\appdata\local\pip\cache\wheels\f8\4e\c3\c8e996b2f75
f503d8686cd9f4eeec04b10d1fc3694363bef13
Successfully built pandoc
Installing collected packages: ply, plumbum, pandoc

   ---------------------------------------- 0/3 [ply]
   ------------- -------------------------- 1/3 [plumbum]
   ------------- -------------------------- 1/3 [plumbum]
   ------------- -------------------------- 1/3 [plumbum]
   ------------- -------------------------- 1/3 [plumbum]
   ------------- -------------------------- 1/3 [plumbum]
   ------------- -------------------------- 1/3 [plumbum]
   ---------------------------------------- 3/3 [pandoc]

Successfully installed pandoc-2.4 plumbum-2.0.2 ply-3.11
```

In [3]:

```
!pip install pandas
```

```
Requirement already satisfied: pandas in C:\Users\dell\AppData\Local\Programs\Python\Pyt
hon314\Lib\site-packages (3.0.5)
Requirement already satisfied: numpy>=2.3.3 in C:\Users\dell\AppData\Local\Programs\Pyth
on\Python314\Lib\site-packages (from pandas) (2.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\dell\AppData\Local\Pro
grams\Python\Python314\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\dell\AppData\Local\Programs\Python\Pyt
hon314\Lib\site-packages (from pandas) (2026.3)
Requirement already satisfied: six>=1.5 in C:\Users\dell\AppData\Local\Programs\Python\P
ython314\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

In [2]:

```
import numpy as np
```

In [3]:

```
import pandas as pd
```

In [4]:

```
df = pd.read_csv(r'C:\Users\dell\AppData\Local\Programs\Python\funnel.csv')
```

In [5]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 542 entries, 0 to 541
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   user_id    542 non-null    str
 1   step       542 non-null    str
 2   timestamp  542 non-null    str
dtypes: str(3)
memory usage: 12.8 KB
```

In [6]:

```
df.describe()
```

Out[6]:

	user_id	step	timestamp
count	542	542	542
unique	200	5	93
top	U055	visited_site	2026-07-20 04:05:00
freq	3	200	17

In [7]:

```
df
```

Out[7]:

	user_id	step	timestamp
0	U055	signup_started	2026-07-20 03:51:00
1	U020	signup_started	2026-07-20 04:06:00
2	U028	visited_site	2026-07-20 03:59:00
3	U096	visited_site	2026-07-20 03:40:00
4	U041	visited_site	2026-07-20 04:01:00
...	...	...	...
537	U094	email_verified	2026-07-20 04:25:00
538	U123	purchase_completed	2026-07-20 04:55:00
539	U019	details_filled	2026-07-20 04:09:00
540	U125	visited_site	2026-07-20 03:55:00
541	U169	details_filled	2026-07-20 04:26:00

542 rows × 3 columns

In [ ]:

```

```

In [15]:

```
!pip install numpy
```

```
Requirement already satisfied: numpy in C:\Users\dell\AppData\Local\Programs\Python\Pyth
on314\Lib\site-packages (2.5.1)
```

In [8]:

```
import numpy as np
```

In [9]:

```
df.count()
```

Out[9]:

```
user_id      542
step         542
timestamp    542
dtype: int64
```

In [20]:

```
df.isnull()
```

Out[20]:

	user_id	step	timestamp
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
...	...	...	...
537	False	False	False
538	False	False	False
539	False	False	False
540	False	False	False
541	False	False	False

542 rows × 3 columns

In [23]:

```
df.isnull().sum()
```

Out[23]:

```
user_id      0
step         0
timestamp    0
dtype: int64
```

In [25]:

```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in C:\Users\dell\AppData\Local\Programs\Python
\Python314\Lib\site-packages (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\dell\AppData\Local\Programs
\Python\Python314\Lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\dell\AppData\Local\Programs\Pyth
on\Python314\Lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\dell\AppData\Local\Programs
\Python\Python314\Lib\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\dell\AppData\Local\Programs
\Python\Python314\Lib\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: numpy>=1.25 in C:\Users\dell\AppData\Local\Programs\Pytho
n\Python314\Lib\site-packages (from matplotlib) (2.5.1)
Requirement already satisfied: packaging>=20.0 in C:\Users\dell\AppData\Local\Programs\P
ython\Python314\Lib\site-packages (from matplotlib) (26.2)
Requirement already satisfied: pillow>=9 in C:\Users\dell\AppData\Local\Programs\Python
\Python314\Lib\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\dell\AppData\Local\Programs\Pyth
on\Python314\Lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\dell\AppData\Local\Progr
ams\Python\Python314\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\dell\AppData\Local\Programs\Python\P
ython314\Lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

In [10]:

```
import matplotlib.pyplot as plt
```

In [47]:

```
df.count()
```

Out[47]:

```
user_id      542
step         542
timestamp    542
dtype: int64
```

In [1]:

```
!pip install datetime
```

```
Collecting datetime
  Downloading datetime-6.0-py3-none-any.whl.metadata (34 kB)
Collecting zope.interface (from datetime)
  Downloading zope_interface-8.5-cp314-cp314-win_amd64.whl.metadata (48 kB)
Collecting pytz (from datetime)
  Downloading pytz-2026.3.post1-py2.py3-none-any.whl.metadata (22 kB)
Downloading datetime-6.0-py3-none-any.whl (52 kB)
Downloading pytz-2026.3.post1-py2.py3-none-any.whl (508 kB)
Downloading zope_interface-8.5-cp314-cp314-win_amd64.whl (215 kB)
Installing collected packages: pytz, zope.interface, datetime

   ---------------------------------------- 0/3 [pytz]
   ---------------------------------------- 0/3 [pytz]
   ---------------------------------------- 0/3 [pytz]
   ---------------------------------------- 0/3 [pytz]
   ---------------------------------------- 0/3 [pytz]
   ------------- -------------------------- 1/3 [zope.interface]
   ------------- -------------------------- 1/3 [zope.interface]
   ------------- -------------------------- 1/3 [zope.interface]
   ------------- -------------------------- 1/3 [zope.interface]
   ------------- -------------------------- 1/3 [zope.interface]
   ------------- -------------------------- 1/3 [zope.interface]
```

```
   ------------------------- ------------- 2/3 [datetime]
   ------------------------- ------------- 2/3 [datetime]
   --------------------------------------- 3/3 [datetime]

Successfully installed datetime-6.0 pytz-2026.3.post1 zope.interface-8.5
```

In [14]:

```python
from datetime import datetime
```

In [12]:

```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 542 entries, 0 to 541
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   user_id    542 non-null    str
 1   step       542 non-null    str
 2   timestamp  542 non-null    str
dtypes: str(3)
memory usage: 12.8 KB
```

In [15]:

```python
df['timestamp'] = pd.to_datetime(df['timestamp'] )
```

In [16]:

```python
df['user_id'] = df['user_id'].astype(object)
```

In [17]:

```python
df['step'] = df['step'].astype(object)
```

In [18]:

```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 542 entries, 0 to 541
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   user_id    542 non-null    object
 1   step       542 non-null    object
 2   timestamp  542 non-null    datetime64[us]
dtypes: datetime64[us](1), object(2)
memory usage: 12.8+ KB
```

In [19]:

```python
df['hours']=df['timestamp'].dt.time
```

In [20]:

```python
df
```

Out[20]:

	user_id	step	timestamp	hours
0	U055	signup_started	2026-07-20 03:51:00	03:51:00
1	U020	signup_started	2026-07-20 04:06:00	04:06:00
2	U028	visited_site	2026-07-20 03:59:00	03:59:00

	user_id	step	timestamp	hours
3	U096	visited_site	2026-07-20 03:40:00	03:40:00
4	U041	visited_site	2026-07-20 04:01:00	04:01:00
...	...	...	...	...
537	U094	email_verified	2026-07-20 04:25:00	04:25:00
538	U123	purchase_completed	2026-07-20 04:55:00	04:55:00
539	U019	details_filled	2026-07-20 04:09:00	04:09:00
540	U125	visited_site	2026-07-20 03:55:00	03:55:00
541	U169	details_filled	2026-07-20 04:26:00	04:26:00

542 rows × 4 columns

In [28]:

```
df
```

Out[28]:

	user_id	step	timestamp	date	hours
0	U055	signup_started	2026-07-20 03:51:00	2026-07-20	03:51:00
1	U020	signup_started	2026-07-20 04:06:00	2026-07-20	04:06:00
2	U028	visited_site	2026-07-20 03:59:00	2026-07-20	03:59:00
3	U096	visited_site	2026-07-20 03:40:00	2026-07-20	03:40:00
4	U041	visited_site	2026-07-20 04:01:00	2026-07-20	04:01:00
...	...	...	...	...	...
537	U094	email_verified	2026-07-20 04:25:00	2026-07-20	04:25:00
538	U123	purchase_completed	2026-07-20 04:55:00	2026-07-20	04:55:00
539	U019	details_filled	2026-07-20 04:09:00	2026-07-20	04:09:00
540	U125	visited_site	2026-07-20 03:55:00	2026-07-20	03:55:00
541	U169	details_filled	2026-07-20 04:26:00	2026-07-20	04:26:00

542 rows × 5 columns

In [21]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 542 entries, 0 to 541
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   user_id    542 non-null    object
 1   step       542 non-null    object
 2   timestamp  542 non-null    datetime64[us]
 3   hours      542 non-null    object
dtypes: datetime64[us](1), object(3)
memory usage: 17.1+ KB
```

In [30]:

```
df['date'] = pd.to_datetime(df['date'] )
```

In [37]:

```
df
```

Out[37]:

	user_id	step	timestamp	date	hours
0	U055	signup_started	2026-07-20 03:51:00	2026-07-20	03:51:00
1	U020	signup_started	2026-07-20 04:06:00	2026-07-20	04:06:00
2	U028	visited_site	2026-07-20 03:59:00	2026-07-20	03:59:00
3	U096	visited_site	2026-07-20 03:40:00	2026-07-20	03:40:00
4	U041	visited_site	2026-07-20 04:01:00	2026-07-20	04:01:00
...	...	...	...	...	...
537	U094	email_verified	2026-07-20 04:25:00	2026-07-20	04:25:00
538	U123	purchase_completed	2026-07-20 04:55:00	2026-07-20	04:55:00
539	U019	details_filled	2026-07-20 04:09:00	2026-07-20	04:09:00
540	U125	visited_site	2026-07-20 03:55:00	2026-07-20	03:55:00
541	U169	details_filled	2026-07-20 04:26:00	2026-07-20	04:26:00

542 rows × 5 columns

In [23]:

```
ab=df.groupby(['step'])['user_id'].count()
```

In [25]:

```
ab
```

Out[25]:

```
step
details_filled         96
email_verified         52
purchase_completed     44
signup_started        150
visited_site          200
Name: user_id, dtype: int64
```

In [26]:

```
df['user_id'].nunique()
```

Out[26]:

```
200
```

total number of users that visited our site is 200

In [27]:

```
import matplotlib.pyplot as plt
```

In [30]:

```
DO = pd.read_csv(r'C:\Users\dell\AppData\Local\Programs\Python\Python314\Scripts\drop of
## impoted a CSV FILE here and that file is extracted using mysql workbence and ordered
## we also can do this task through pyhton
```

In [31]:

```
DO
```

Out[31]:

	step	count
0	visited_site	200
1	signup_started	150
2	details_filled	96
3	email_verified	52
4	purchase_completed	44

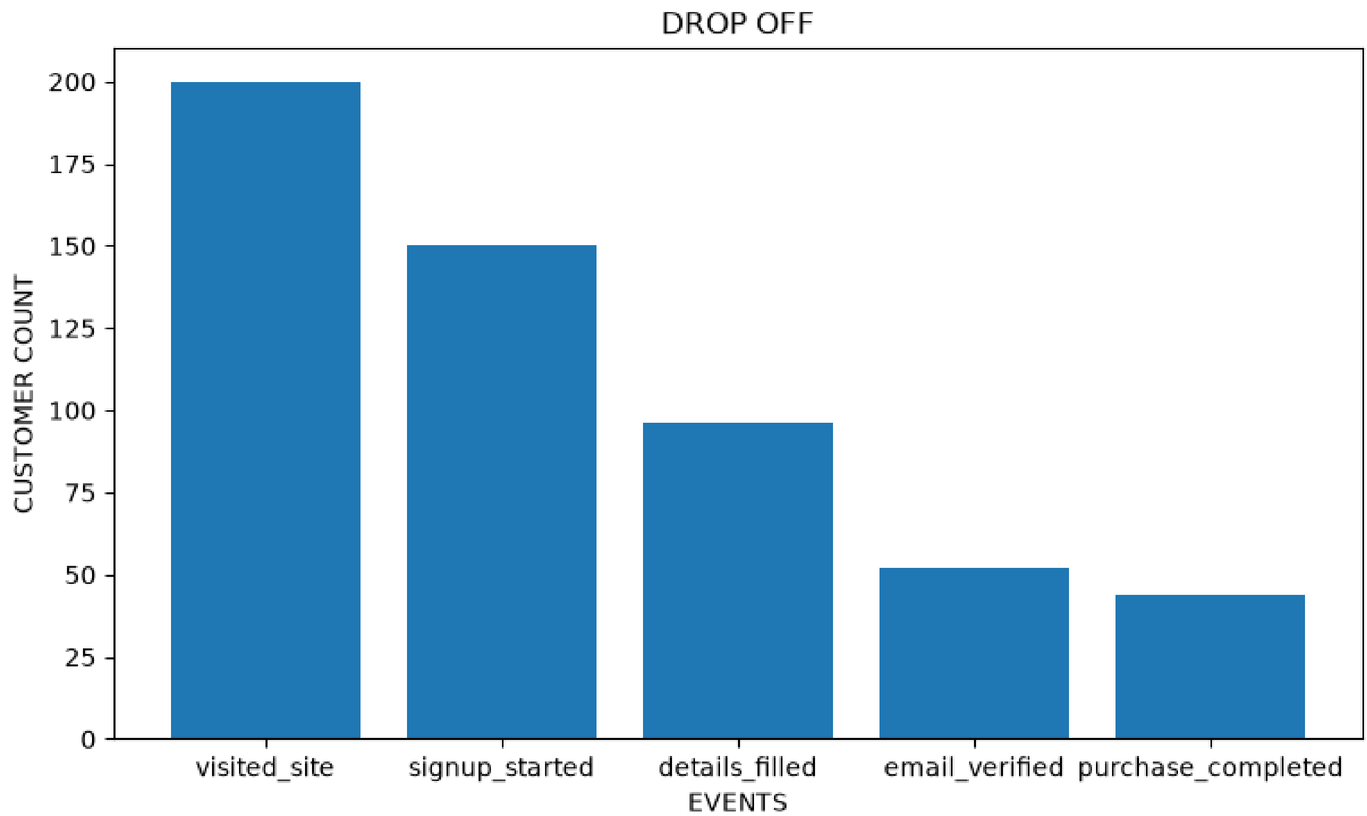
In [37]:

```
plt.figure(figsize=(9,5))

plt.bar(DO['step'],DO['count'])
plt.title('DROP OFF')
plt.xlabel('EVENTS')
plt.ylabel('CUSTOMER COUNT')
```

Out[37]:

```
Text(0, 0.5, 'CUSTOMER COUNT')
```



i've noticed the biggest drop off is at signup_started --> details filled is 54

In [38]:

```
DO['conversion_rate_perc'] =( DO['count']/200)*100
```

In [39]:

```
DO
```

Out[39]:

	step	count	conversion_rate_perc
0	visited_site	200	100.0
1	signup_started	150	75.0
2	details_filled	96	48.0
3	email_verified	52	26.0
4	purchase_completed	44	22.0

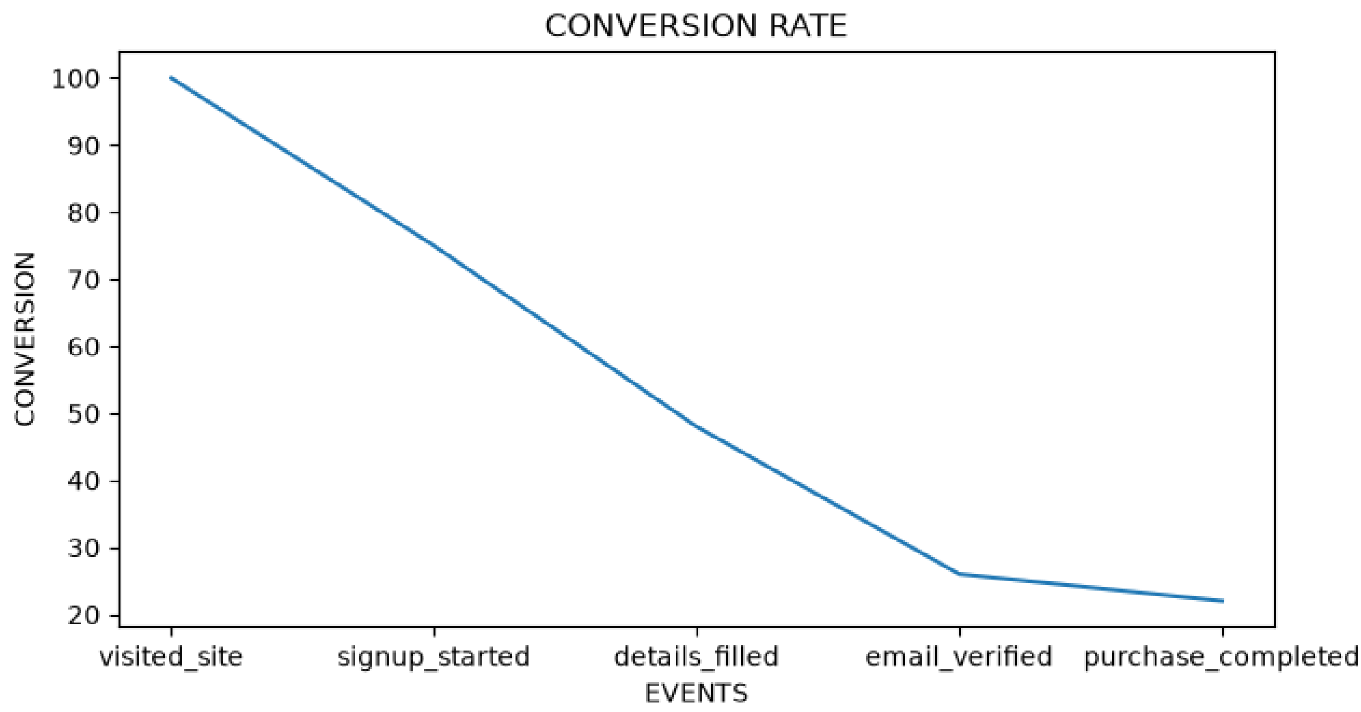
In [42]:

```
plt.figure(figsize=(8,4))

plt.plot(DO['step'],DO['conversion_rate_perc'])
plt.title('CONVERSION RATE')
plt.xlabel('EVENTS')
plt.ylabel('CONVERSION')
```

Out[42]:

```
Text(0, 0.5, 'CONVERSION')
```



REPORT

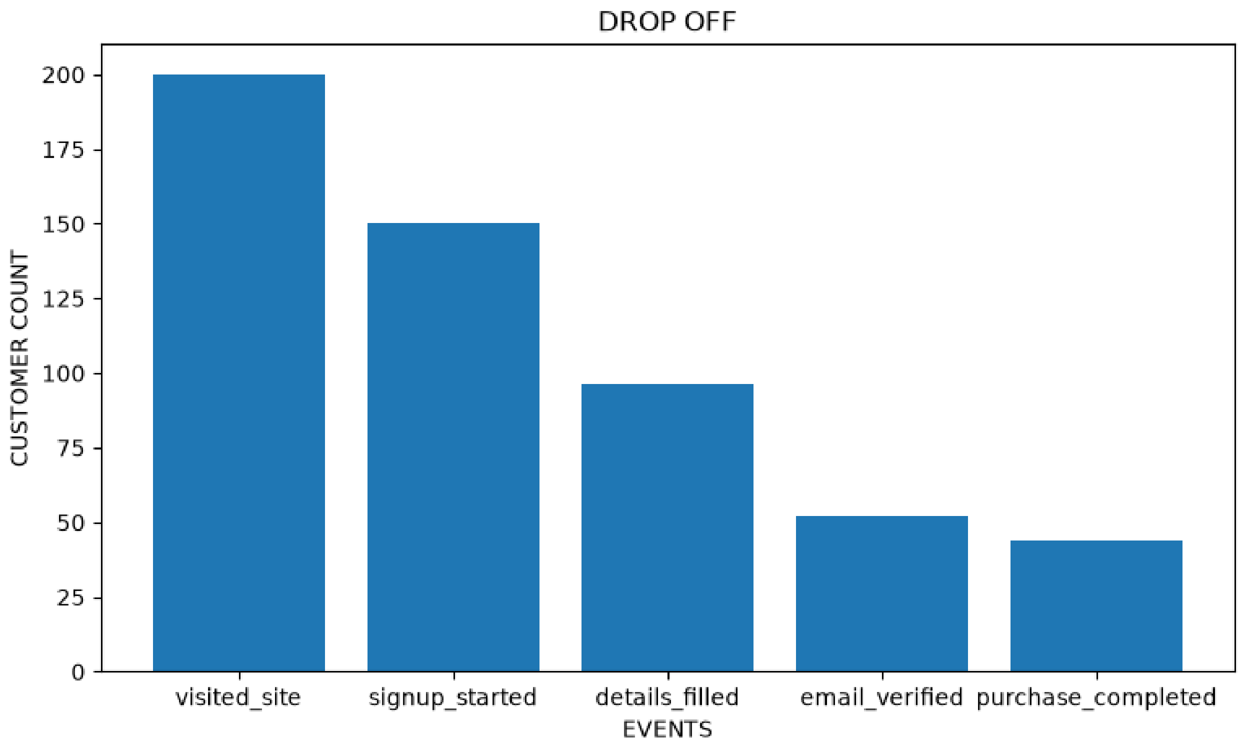
total records --> 542

total uniuqe customers --> 200

customers_count at each stage

	step	count
0	visited_site	200
1	signup_started	150
2	details_filled	96
3	email_verified	52
4	purchase_completed	44

barchart of dropoff



i've noticed the biggest drop off at signup_started --> details filled and that is 54

reasons for dropoff

asking for un unecessary data

slow page loading and page errors

tiny input fields or missing autofill functionality

how to fix it

ask only needed data

add one click registration via google etc.,

show a progress bar that shows users how many steps are left

add social proof or trust badges so that we can reduce the anxiety about sharing data

conversion rate

	step	count	conversion_rate_perc
0	visited_site	200	100.0
1	signup_started	150	75.0
2	details_filled	96	48.0
3	email_verified	52	26.0
4	purchase_completed	44	22.0

note: conversion_rate = customers entering an event divided by total customeres *100

In [ ]: